

Stabilizers of smooth Banach–Lie actions are Lie subgroups

Autonomous solution packet for arXiv:math/0703710
Model: GPT5.6

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Abstract

An–Neeb ask whether the stabilizer of a point under an arbitrary smooth action of a Banach–Lie group on a Banach manifold is always a Banach–Lie subgroup. We prove that it is; in fact a C^2 action suffices. The key is a uniform no-small-period lemma. In a chart at the fixed point, the vector field generated by $x \in \mathfrak{g}$ has local Lipschitz constant $O(\|x\|)$. If $\exp(x)$ fixes the point and x is small, Gronwall’s inequality forces the initial velocity to vanish. Flow uniqueness then makes the whole one-parameter group fix the point. Consequently the stabilizer is, in an exponential chart, exactly its closed infinitesimal stabilizer algebra.

1 Source problem and answer

The source proves the conclusion when the tangent space to the orbit is closed and then asks the unrestricted question [1, Problem 4.5]:

Problem 4.5 Suppose that the Banach–Lie group G acts smoothly on the Banach manifold M and $p \in M$. Is it always true that the stabilizer G_p is a Lie subgroup of G ?

Here “Lie subgroup” means a Banach–Lie group in the subspace topology, without requiring its Lie algebra to be complemented in $\mathfrak{g}$.

Theorem 1 (affirmative answer). *Let G be a Banach–Lie group acting by a C^2 action*

$$\sigma : G \times M \longrightarrow M$$

on a Banach C^2 -manifold M . For every $p \in M$, the stabilizer

$$G_p = \{g \in G : g.p = p\}$$

is a Banach–Lie subgroup of G with its subspace topology.

Thus Problem 4.5 has a positive answer, with weaker regularity than the smoothness assumed there. No closed-range or complementability hypothesis is needed.

2 Proof intuition

Write x_M for the fundamental vector field generated by $x \in \mathfrak{g}$. Near p , its chart expression $F(x, y)$ is linear in x . Its derivative in the y -variable is consequently bounded by $C\|x\|$, uniformly near p . The orbit $t \mapsto \exp(tx).p$ solves $y' = F(x, y)$. If it returns to p at time one, comparison with the constant vector $F(x, 0)$ shows that

$$\|F(x, 0)\| \leq C\|x\|e^{C\|x\|}\|F(x, 0)\|.$$

For small x the right coefficient is less than one, so the initial velocity vanishes. The orbit is then constant by ODE uniqueness. This converts a time-one stabilizer relation into an infinitesimal stabilizer relation, uniformly in every sufficiently small logarithm.

3 A parametric no-small-return lemma

Lemma 2. *Let P and E be Banach spaces, let $V \subseteq E$ be a neighborhood of 0, let $A : V \rightarrow \mathcal{L}(P, E)$ be C^1 , and put $F(x, y) = A(y)x$. For x near 0, suppose $y_x : [0, 1] \rightarrow V$ solves*

$$y'_x(t) = F(x, y_x(t)), \quad y_x(0) = 0,$$

and that all these curves remain in one fixed neighborhood of 0 in V . Then there is a neighborhood U of 0 in P such that

$$x \in U, \quad y_x(1) = 0 \quad \implies \quad F(x, 0) = 0.$$

In that event $y_x(t) = 0$ for every $t \in [0, 1]$.

Proof. The derivative of A gives a continuous family of bounded bilinear maps

$$B(y)(x, z) := (DA(y)z)x.$$

After shrinking the fixed y -neighborhood, there is $C > 0$ such that $\|B(y)\| \leq C$ throughout it. The mean-value formula on a convex ball therefore gives

$$\|F(x, y) - F(x, 0)\| \leq C\|x\| \|y\|. \tag{1}$$

Put $v = F(x, 0)$ and $\lambda = C\|x\|$. Integrating the ODE and using (1) gives, for $0 \leq t \leq 1$,

$$\|y_x(t)\| \leq t\|v\| + \lambda \int_0^t \|y_x(s)\| ds.$$

Gronwall's inequality yields

$$\|y_x(t)\| \leq te^{\lambda t}\|v\|. \tag{2}$$

If $y_x(1) = 0$, another integration of the ODE gives

$$v = - \int_0^1 (F(x, y_x(s)) - F(x, 0)) ds.$$

Combining this with (1) and (2),

$$\|v\| \leq \lambda \int_0^1 se^{\lambda s} ds \|v\| \leq \lambda e^{\lambda} \|v\|.$$

Shrink U so that $\lambda e^{\lambda} < 1$ there. Then $v = 0$. The constant curve is now a solution with the same initial value, so uniqueness for Banach space ODEs gives $y_x \equiv 0$. $\square$

4 Proof of the theorem

Proof of Theorem 1. Fix $p \in M$ and put $H = G_p$. It is a closed subgroup. Let

$$\alpha_p : \mathfrak{g} \longrightarrow T_p M, \quad \alpha_p(x) = \left. \frac{d}{dt} \right|_{t=0} \exp_G(tx).p,$$

and set $\mathfrak{h} = \ker \alpha_p$. This is a closed linear subspace. It is also a Lie subalgebra: the fundamental vector fields generated by elements of $\mathfrak{h}$ vanish at p , and hence so does their bracket.

Choose a chart $\kappa : W \rightarrow V \subseteq E$ with $\kappa(p) = 0$. The chart expression of the fundamental fields is

$$F(x, y) = T_{\kappa^{-1}(y)} \kappa \left(\left. \frac{d}{dt} \right|_{t=0} \exp_G(tx). \kappa^{-1}(y) \right).$$

It has the form $F(x, y) = A(y)x$ for a C^1 map $A : V \rightarrow \mathcal{L}(\mathfrak{g}, E)$: the C^2 regularity of the action supplies precisely this C^1 dependence of its first differential. Thus Lemma 2 applies.

For all sufficiently small $x \in \mathfrak{g}$, the curve

$$y_x(t) = \kappa(\exp_G(tx).p), \quad 0 \leq t \leq 1,$$

stays in one fixed chart neighborhood. This follows uniformly in t from continuity of $(t, x) \mapsto \exp_G(tx).p$ and compactness of $[0, 1]$. It solves the ODE of Lemma 2, with $F(x, 0) = T_p \kappa(\alpha_p(x))$.

Take a sufficiently small open 0-neighborhood $U \subseteq \mathfrak{g}$ on which $\exp_G$ is a diffeomorphism onto an identity neighborhood and on which the lemma applies. If $x \in U$ and $\exp_G(x) \in H$, then $y_x(1) = 0$, so the lemma implies $\alpha_p(x) = 0$, i.e. $x \in \mathfrak{h}$. Conversely, if $x \in \mathfrak{h}$, the orbit through p and the constant curve through p solve the same autonomous ODE with the same initial value. Uniqueness gives $\exp_G(tx).p = p$ for all $t \in \mathbb{R}$, and in particular $\exp_G(x) \in H$. We have proved the local identity

$$H \cap \exp_G(U) = \exp_G(U \cap \mathfrak{h}). \quad (3)$$

Since $\mathfrak{h}$ is closed, it is a Banach space. Equation (3) makes $\log_G$ a chart from the open identity neighborhood $H \cap \exp_G(U)$ in the subspace topology onto $U \cap \mathfrak{h}$. Its left translates form a Banach-manifold atlas on H . The chart expressions of multiplication and inversion are restrictions of the smooth local group operations of G (equivalently, of the local BCH operations), so they are smooth. Thus H is a Banach–Lie group, and its inclusion into G is a smooth topological embedding. Hence it is a Banach–Lie subgroup in the source’s sense. $\square$

Remark 3 (what is and is not solved). *The theorem removes the closed-orbit-tangent assumption in An–Neeb’s Theorem 4.2 and fully answers their Problem 4.5. It does not construct a Banach-manifold structure on G/H for an arbitrary nonsplit Banach–Lie subgroup H , so the opposite direction of their Conjecture 4.1 remains outside the claim.*

5 Literature and novelty boundary

Jotz–Neeb prove the proper-action case by showing that every orbit tangent has closed range [2]; combined with An–Neeb’s theorem, this yields Lie stabilizers for proper actions. Diez–Rudolph still state that the general stabilizer question is unknown even for Banach–Lie actions [3]. Most decisively, Glöckner–Neeb’s 2026 monograph manuscript repeats the exact Banach transformation-group problem and the same known closed-range and linear-action subcases [4].

The run searched its cheap indexes and performed bounded arXiv searches on 2026-08-11 using the exact problem sentence and combinations of “stabilizer”, “Banach–Lie subgroup”, “smooth action”, and “Problem IX.3.b”. These searches returned the source, the proper-action paper, the 2022 restatement, and the 2026 monograph, but no unrestricted answer. The conservative status is therefore a candidate full solution pending human verification. Review should focus on the uniform mixed- derivative estimate in Lemma 2; it is the step that rules out the natural weighted-rotation counterexample attempts.

References

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